

AHMED IQDYMAT

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Automation & Control Systems Engineer

SUMMARY

Automation & Control Systems Engineer with hands-on industrial experience in production, CNC equipment, maintenance, and troubleshooting, combined with expertise in classical and intelligent control, system modelling, simulation, testing, and validation. Proficient in MATLAB/Simulink, with experience in ROS 2, reinforcement learning, and industrial robotics. Currently completing a PhD in Systems Engineering and seeking an engineering role in industrial automation, control systems, systems engineering, simulation, testing, validation, or R&D.

EXPERIENCE

Dec 2021 - Present	Automation & Control Systems Engineer — PhD Research National University of Science and Technology POLITEHNICA Bucharest, Romania - Developed and tested control and simulation workflows for industrial automation and robotic systems using MATLAB/Simulink. - Designed and evaluated control architectures using classical control and reinforcement learning methods. - Integrated MATLAB/Simulink with ROS 2 and URSim for closed-loop simulation, testing, and performance evaluation. - Worked with UR3e and ABB IRB120 industrial robots in simulation environments. - Analysed system behaviour, controller performance, trajectory accuracy, and test results; prepared technical documentation, reports, and presentations.
Jun 2021 - Nov 2021	Industrial Automation & Maintenance Engineer TB Lock - Safety Door Manufacturing Hebron, West Bank - Coordinated production activities and supported daily manufacturing operations. - Operated and monitored CNC and automated manufacturing equipment. - Performed preventive and corrective maintenance, equipment troubleshooting, and fault diagnosis. - Supported the monitoring and adjustment of machine operating parameters to maintain stable production performance. - Monitored production processes and product quality to support reliable and efficient manufacturing.
Jan 2019 - Dec 2020	Operation and Maintenance Engineer Al Jebrini Dairy & Food Industries Hebron, West Bank - Operated and monitored industrial production machinery and production lines. - Performed routine maintenance, corrective maintenance, and equipment troubleshooting. - Supported production continuity, equipment availability, and safe machine operation. - Followed safety, hygiene, and quality procedures in a food manufacturing environment.

EDUCATION

2021 - Present	PhD Candidate in Systems Engineering National University of Science and Technology POLITEHNICA Bucharest Romania Focus: Systems engineering, automation, control systems, system modelling and simulation, intelligent control, reinforcement learning, and industrial robotics.
2016 - 2018	Master's Degree in Automation and Industrial Informatics Djilali Bounaama University of Khemis Miliana Algeria
2011 - 2016	Bachelor's Degree in Automation Djilali Bounaama University of Khemis Miliana Algeria

SKILLS

Automation & Control: Industrial Automation, Control Systems, Classical Control, Process Control, System Modelling, Intelligent Control	
Engineering & Simulation: MATLAB, Simulink, System Simulation, Performance Analysis, Testing & Validation, Engineering Analysis	
Industrial Engineering & Maintenance: CNC Operation, Machine Operation, Preventive & Corrective Maintenance, Troubleshooting, Production Support, Equipment Monitoring	
Robotics & Intelligent Systems: ROS 2, Industrial Robotics, Reinforcement Learning, LQR-Based Control, Trajectory Planning	
Systems Engineering & Research: Systems Engineering, Experimental Evaluation, Technical Documentation, Research Methodology, Scientific Writing, Technical Presentations	
Programming & Data: MATLAB Scripting, Python (Basic), Data Analysis	

LANGUAGES

ARABIC Native
ENGLISH Advanced
FRENCH Intermediate
ROMANIAN Elementary

CERTIFICATIONS & COURSES

ICDL - International Computer Driving Licence

PUBLICATIONS

- "Sim-to-Sim Control of a UR3e Robot Using Soft Actor-Critic and Real-Time MATLABBROSURSim Integration." IEEE IDAACS, 2025.
- "Reinforcement Learning of a Six-DOF Industrial Manipulator for Pick-and-Place Application Using Efficient Control in Warehouse Management." Sustainability, 2025.
- "Analysis of Trajectory Optimization for Solving the Tower of Hanoi Problem Using an Industrial Manipulator." IEEE CSCS, 2023.

RESEARCH PROFILES

- orcid.org/0009-0009-8078-8910
- scholar.google.com/citations?user=hNu7HPkAAAAJ